

TAD WeightedGraph			
WeightedGraph = (V, A)			
$\forall (u, v) \in V \times V :$ $((u, v, w) \in A \Leftrightarrow \text{exists a directed edge from } u \text{ to } v \text{ with weight } w)$			
Primitive Operations			
Operation	Input	Input	Type
addNode	T	void	Modifier
addEdge	T, T, int	void	Modifier
getNodes	—	List<T>	Observer
getEdges	—	List<Edge>	Observer
getNeighbors	T	List<T>	Observer
getEdgeWeight	T, T	int	Observer
hasEdge	T, T	boolean	Observer
getNeighborsWithWeights	T	List<Pair>	Observer

$\langle \text{addNode}(T \text{ value}) \rangle$
Adds a new node to the graph if it doesn't already exists
Precondition: $\text{node} \neq \text{null} \wedge \text{node} \notin V$
Postconditon: $V \leftarrow V \cup \{n\}$

$\langle \text{addEdge}(T \text{ from}, T \text{ to}, \text{int weight}) \rangle$
Adds a new directed edge with weight from node to another
$\text{from} \in V \wedge \text{to} \in V \wedge \text{weight} \geq 0$
$A \leftarrow A \cup \{(\text{from}, \text{to}, \text{weight})\}$

$\langle \text{getNodes}() \rangle$
Returns the list of nodes in the graph.
Precondition: —
Postcondition: A list L where: $\forall x \in L, x \in V.$

$\langle \text{getEdges}() \rangle$
Returns the list of edges in the graph.
Precondition: —
Postcondition: A list L where: $\forall (u, v, w) \in E \Rightarrow (u, v, w) \in A.$

$\langle \text{getNeighbors}(T \text{ nodeId}) \rangle$

Returns the adjacent neighbors of a node.
Precondition: $nodeId \in V$
Postcondition: returns $L = \{v \in V \mid \exists w : (nodeId, v, w) \in A\}$
$\langle getEdgeWeight(T \text{ from}, T \text{ to}) \rangle$
Returns the weight from a node to another.
Precondition: $from \in V \wedge to \in V$
Postcondition: if $(from, to, w) \in A$, returns w , otherwise returns -1 .
$\langle hasEdge(T \text{ from}, T \text{ to}) \rangle$
Verifies if an edge from a node to another exists.
Precondition: $from \in V \wedge to \in V$
Postcondition: returns $true \Leftrightarrow \exists w : (from, to, w) \in A$
$\langle getNeighborsWithWeights(T \text{ nodeId}) \rangle$
Returns a node's neighbors along with the edge weight to them
Precondition: $nodeId \in V$
Postcondition: Returns $L = \{ (v, w) \mid (nodeId, v, w) \in A \}$